

Incomplete Exam for the First Midterm

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1. Rewrite the integral using the given substitution.
(4 Points)

$$\int x\sqrt{4+2x+x^2} \, dx, \quad u = 1+x$$

- ☐ $\int (u-1)\sqrt{u^2+3} \, du$ ✓
- ☐ $\int (u-1)\sqrt{u^2-3} \, du$
- ☐ $\int u\sqrt{u^2+3} \, du$
- ☐ $\int (u-1)(u^2+3) \, du$
- ☐ $\int (u-1)^{1/2}(u^2+3) \, du$

2. Evaluate
(4 Points)

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x - (x^2/2)}{x^3} =$$

- ☐ $\frac{1}{6}$ ✓
- ☐ $-\frac{1}{6}$
- ☐ $\frac{1}{3}$
- ☐ $-\frac{1}{3}$
- ☐ -3

3. Which of the following is a bounded divergent sequence?
(3 Points)

- ☐ $a_n = (-1)^n$ ✓
- ☐ $a_n = \frac{1}{n}$
- ☐ $a_n = e^n$
- ☐ $a_n = n$
- ☐ None of these given choices

4. Which of the following is correct about the sequence?
(4 Points)

$$\{a_n\}_{n=1}^{\infty} = \{ \ln(\sin(\frac{1}{n})) \}_{n=1}^{\infty}$$

- ☐ It is convergent with the limit, $\ln 3$ ✓
- ☐ Divergent
- ☐ It is convergent with the limit, e^3
- ☐ It is convergent with the limit, 3
- ☐ It is convergent with the limit, $\ln \frac{1}{3}$

5. What is the correct form of the partial fraction decomposition of the following function?
(4 Points)

$$\frac{3x-2}{x^2(x^2+1)(x^4+1)}$$

- ☐ $\frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{D}{x^2-1} + \frac{E}{x^2} + \frac{Gx+H}{(x^2+1)^2}$
- ☐ $\frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{D}{(x-1)^2} + \frac{E}{x} + \frac{Gx+H}{(x^2+1)^2}$
- ☐ $\frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{D}{x-1} + \frac{E}{x} + \frac{Gx+H}{(x^2+1)^2}$
- ☐ $\frac{A}{x+3} + \frac{Bx+C}{x^2+1} + \frac{D}{x-3} + \frac{Ex+F}{x+3}$
- ☐ $\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1} + \frac{D}{x+1} + \frac{Ex+F}{x^2+1} + \frac{Gx+H}{(x^2+1)^2}$ ✓

6. Calculate
(4 Points)

$$\int \frac{\sqrt{2x-1}}{2x+3} \, dx =$$

- ☐ $\sqrt{2x-1} - 2 \arctan\left(\frac{\sqrt{2x-1}}{2x}\right) + C$
- ☐ $\sqrt{2x-1} - \arctan\left(\frac{\sqrt{2x+2}}{2}\right) + C$
- ☐ $(2x-1) - 2 \arctan\left(\frac{\sqrt{2x-1}}{2}\right) + C$
- ☐ $(2x-1)^{1/3} - 2 \arctan\left(\frac{\sqrt{2x-1}}{2}\right) + C$
- ☐ $(2x-1)^{1/2} - 2 \arctan\left(\frac{\sqrt{2x-1}}{2}\right) + C$ ✓

7. Which of the following is correct for the given integral?
(4 Points)

$$I = \int_0^1 (x^4 + 1)e^{-x} \, dx$$

- ☐ $I = -6e^{-1} + 1 + 12 \int_0^1 x^2 e^{-x} \, dx$ ✓
- ☐ $I = -6e^{-1} + 12 + 12 \int_0^1 x e^{-x} \, dx$
- ☐ $I = -6e^{-1} + 1 + 12 \int_0^1 x^3 e^{-x} \, dx$
- ☐ $I = -2e^{-1} - 1 + 2 \int_0^1 e^{-x} \, dx$
- ☐ None of these given choices

8. Calculate
(4 Points)

$$\int \frac{2}{(x^2 + 2x + 2)^2} \, dx =$$

- ☐ $\frac{x+1}{x^2+2x+2} + \arctan(x+1) + C$ ✓
- ☐ $2 \frac{x+1}{x^2+2x+2} + 2 \arctan(x+1) + C$
- ☐ $\frac{\sin(2(x+1))}{x^2+2x+2} + \arctan(x+1) + C$
- ☐ $\frac{\sin(2 \arctan(x+1))}{2} + \tan(x+1) + C$
- ☐ $\frac{\sin(2 \tan(x+1))}{2} - \arctan(x+1) + C$

9. Rewrite the integral using the given substitution.
(4 Points)

$$\int \frac{1}{x^2 \sqrt{4x+1}} \, dx, \quad u = \sqrt{4x+1}$$

- ☐ $8 \int \left[\frac{A}{u-1} + \frac{B}{u+1} + \frac{C}{(u+1)^2} + \frac{D}{(u-1)^2} \right] du$ ✓
- ☐ $8 \int \left[\frac{A}{u-1} + \frac{B}{u+1} + \frac{C}{(u+1)^2} + \frac{D}{(u-1)} \right] du$
- ☐ $8 \int \left[\frac{A}{u-1} + \frac{B}{u+1} + \frac{Cu}{(u+1)} + \frac{D}{(u-1)^2} \right] du$
- ☐ $4 \int \left[\frac{A}{u-1} + \frac{B}{u+1} + \frac{C}{(u+1)^2} \right] du$
- ☐ $8 \int \left[\frac{A}{u-1} + \frac{B}{u+1} + \frac{Cu}{(u+1)^2} + \frac{Du}{(u-1)^2} \right] du$

10. Assuming that the following pattern continues, find a general term for
(4 Points)

$$\left\{ -1, \frac{8}{9}, -\frac{12}{16}, \frac{16}{25}, \dots \right\}$$

- ☐ $a_n = (-1)^n \frac{4n}{(n+1)^2}$ ✓
- ☐ $a_n = (-1)^n \frac{4n}{4n-2}$
- ☐ $a_n = (-1)^n \frac{4n}{4n+1}$
- ☐ $a_n = (-1)^{n+1} \frac{n}{2n+2}$
- ☐ $a_n = (-1)^{n+1} \frac{n}{(n+2)^2}$

11. Find the value of the limit for the sequence given.
(3 Points)

$$\left\{ \frac{1.9, 17, \dots, (7n+1)}{7n^2} \right\}$$

- ☐ 0 ✓
- ☐ 7
- ☐ 1
- ☐ -7
- ☐ Divergent

12. Which of the following sequence is decreasing?
(4 Points)

- ☐ $a_n = n \left(\frac{1}{e} \right)^n$ ✓
- ☐ $a_n = n \left(\frac{1}{e} \right)^{\frac{1}{n}}$
- ☐ $a_n = n e^n$
- ☐ $a_n = (-1)^n$
- ☐ None of these given choices

13. Evaluate
(4 Points)

$$\int \cos(x-1) \cdot \sin(x+4) \, dx =$$

- ☐ $\frac{x}{2} \sin(5) - \frac{\cos(2x+3)}{4} + C$ ✓
- ☐ $\frac{\sin(5x)}{2} + \frac{\cos(2x+3)}{4} + C$
- ☐ $\frac{\sin(5)}{4} - \frac{\cos(2x+3)}{4} + C$
- ☐ $\frac{\sin(2-x)}{2} - \frac{\sin(7x)}{14} + C$
- ☐ $\frac{x \sin(5)}{2} - \frac{\cos(7x)}{14} + C$

14. Which of the following is true about the given integral?
(4 Points)

$$I = \int_0^{\infty} \frac{dx}{\sqrt{x}(x+1)}$$

- ☐ $I = \pi$ ✓
- ☐ $I = -\frac{\pi}{2}$
- ☐ $I = \frac{\pi}{2}$
- ☐ I is divergent
- ☐ $I = 0$

15. Suppose that
(3 Points)

$$\int_1^e (\ln t)^3 \, dt = 6 - 2e, \quad \text{and find } I = \int_1^e (\ln t)^4 \, dt$$

- ☐ $I = 9e - 24$ ✓
- ☐ $I = e - 24$
- ☐ $I = 6e - 24$
- ☐ $I = e - 12$
- ☐ $I = 9e - 32$

16. Rewrite the integral using the given substitution.
(4 Points)

$$\int \cot^5 x \sin^4 x \, dx, \quad u = \sin x$$

- ☐ $\int \left(\frac{1}{u} - 2u + u^3 \right) du$ ✓
- ☐ $\int \left(\frac{1}{u^2} - 2u - u^3 \right) du$
- ☐ $\int \left(\frac{1}{u} - 2u^2 + u \right) du$
- ☐ $\int \left(\frac{1}{u} + u + u^4 \right) du$
- ☐ None of these given choices

17. Determine whether the improper integral converges or diverges, and if it converges, find its value.
(4 Points)

$$I = \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{\cos x}{\sqrt{1 - \sin x}} \, dx$$

- ☐ $I = -\sqrt{2}$ ✓
- ☐ $I = 2\sqrt{2}$
- ☐ $I = 2$
- ☐ $I = 4\sqrt{2}$
- ☐ I is divergent

18. Evaluate
(4 Points)

$$I = \int \frac{1}{2 \sin x + \sin 2x} \, dx$$

- ☐ $I = \frac{1}{4} \ln \left| \tan\left(\frac{x}{2}\right) \right| + \frac{1}{8} \tan^2\left(\frac{x}{2}\right) + C$ ✓
- ☐ $I = \frac{1}{2} \ln \left| \tan\left(\frac{x}{4}\right) \right| + \tan^2\left(\frac{x}{2}\right) + C$
- ☐ $I = \frac{1}{4} \ln \left| \arctan\left(\frac{x}{2}\right) \right| + \frac{1}{8} \arctan^2\left(\frac{x}{2}\right) + C$
- ☐ $I = \frac{1}{4} \ln |\tan(x)| + \frac{1}{8} \tan^2(x) + C$
- ☐ $I = 4 \ln |\tan(x)| + 8 \tan^2(x) + C$

19. Evaluate
(4 Points)

$$\lim_{x \rightarrow 0^+} (\cos x)^{1/x^2}$$

- ☐ $\frac{1}{\sqrt{e}}$ ✓
- ☐ $\frac{1}{e^2}$
- ☐ $\frac{1}{e}$
- ☐ $\frac{1}{2e}$
- ☐ $-\frac{1}{e^2}$

20. Rewrite the integral using the given substitution (with an appropriate domain).
(4 Points)

$$\int \frac{1}{x^2 \sqrt{16x^2 - 9}} \, dx, \quad x = \frac{3}{4} \sec u$$

- ☐ $\frac{4}{9} \int \cos u \, du$ ✓
- ☐ $\frac{4}{9} \int \cos 2u \, du$
- ☐ $\frac{4}{3} \int \cos u \, du$
- ☐ $\frac{4}{9} \int \cos^3 u \, du$
- ☐ None of these given choices

21. Which of the following is true about the given integral using the appropriate substitution.
(3 Points)

$$I = \int_0^{\frac{\pi}{6}} 2x \tan(x^2 + 1) \, dx$$

- ☐ $I = [\ln \sec u]_1^{\frac{1}{2}}$ ✓
- ☐ $I = [\ln \sec u]_{\frac{1}{2}}^1$
- ☐ $I = \left[\ln \left| \sec \frac{1}{u} \right| \right]_1^{\frac{1}{2}}$
- ☐ None of these given choices
- ☐ $I = \left[\ln \left| \csc \frac{1}{u} \right| \right]_1^{\frac{1}{2}}$

22. Evaluate
(4 Points)

$$\int_0^{\frac{\pi}{2}} \tan^5 x \sec^4 x \, dx =$$

- ☐ $\frac{117}{8}$ ✓
- ☐ $\frac{121}{8}$
- ☐ $\frac{117}{7}$
- ☐ $\frac{115}{8}$
- ☐ $\frac{127}{8}$

23. Suppose that $f(3)=4$, $f'(1)=1$ and derivative of f is continuous. Find the value of
(4 Points)

$$I = \int_1^3 \frac{x \cdot f'(x) - f(x)}{x^2} \, dx$$

- ☐ $I = \frac{1}{3}$ ✓
- ☐ $I = \frac{1}{2}$
- ☐ $I = \frac{1}{4}$
- ☐ $I = 1$
- ☐ $I = -1$

24. Rewrite the integral using the given substitution.
(4 Points)

$$\int \sin^3 x \sqrt{\cos x} \, dx, \quad u = \cos x$$

- ☐ $\int (u^{5/2} - u^{1/2}) \, du$ ✓
- ☐ $\int (u^{7/2} - u^{3/2}) \, du$
- ☐ $\int (u-1) u^{3/2} \, du$
- ☐ $\int (u^{5/2} + 1) u^2 \, du$
- ☐ $\int (u^2 - 1)^2 \, du$

25. Evaluate
(4 Points)

$$I = \int \frac{x^3}{\sqrt{x^2+9}} \, dx$$

- ☐ $I = \frac{1}{3} (x^2 - 18) \sqrt{x^2 + 9} + C$ ✓
- ☐ $I = \frac{1}{3} (x^2 + 9) - \sqrt{x^2 + 9} + C$
- ☐ $I = \frac{x}{3} (x^2 + 9) - \sqrt{x^2 + 9} + C$
- ☐ $I = \frac{x^2}{3} (x^2 - 18) \sqrt{x^2 + 9} + C$
- ☐ None of these given choices

26. Which of the following is correct for the given integral?
(4 Points)

$$I = \int_0^1 x^2 e^{-x^2} \, dx$$

- ☐ $I = \frac{1}{2} \int_0^{\infty} e^{-x^2} \, dx$ ✓
- ☐ I is divergent
- ☐ $I = \int_0^{\infty} \frac{x^2}{(2e)^{x^2}} \, dx$
- ☐ $I = \int_0^{\infty} \frac{x^2}{(3e)^{x^2}} \, dx$
- ☐ $I = \frac{1}{3} \int_0^{\infty} \frac{x}{(e)^{x^2}} \, dx$